BrightNest

Grounded after-hours help for parents

COMP6708 Mini Demo · **Group 15**

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seanfontaine.dev/COMP6708_A3_Demo

One sentence

1

A learning-centre assistant that

answers from centre truth,
acts with tools,
and hands off when humans must.

Training-free · RAG · hybrid routing · live on the public web

The problem

1

Imagine you run a tutoring centre.

- Parents message after hours: typhoons, fees, seats, trials
- Staff cannot live on WhatsApp 24/7
- A wrong fee or hours answer costs trust — fast

So you hired Group 15

We built **BrightNest**: grounded answers, structured actions, honest hand-offs.

BrightNest

Not a chatbot that invents tuition from thin air.

Technique: training-free RAG + tools

1

Why training-free? Centre docs change weekly ⇒ fine-tuning is the wrong loop.

RAG (self-built)

1. Chunk markdown KB (`content/`)
2. Embed with DashScope
3. Store / query in Chroma
4. Channel pack (3 dense + 2 BM25) → top-k
5. Generate with Qwen + citations

Retrieval $s(q, c) = \mathrm{sim}(E(q), E(c))$ · inspector shows rank, s, chunk.

Tool cascade (not LLM-first)

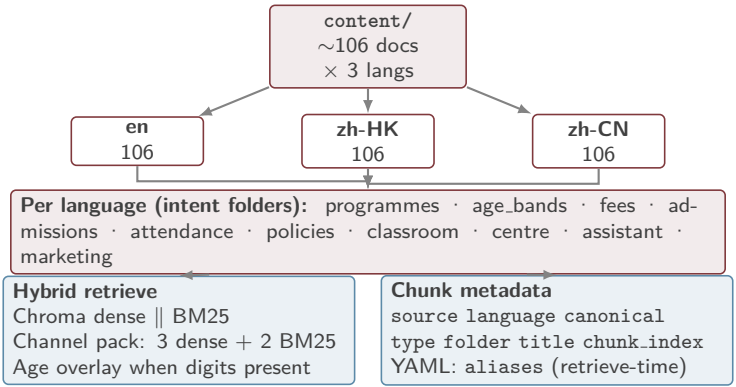
- Registry of tools; L1 detectors pick in <1 ms
- L2 cheap policy only on ambiguous RAG
- UI: **Tool / dialogue** · **Live info** · **RAG**
- Hard FSM + **final check** before commits
- Fees / programmes = pure RAG

cascade not agent-loop

KB may enrich

Knowledge base structure

1



Tool facts (typhoon signals) sit in `backend/facts/severe_weather.json`; grounded compose may still enrich from KB.

Three answer tiers

1

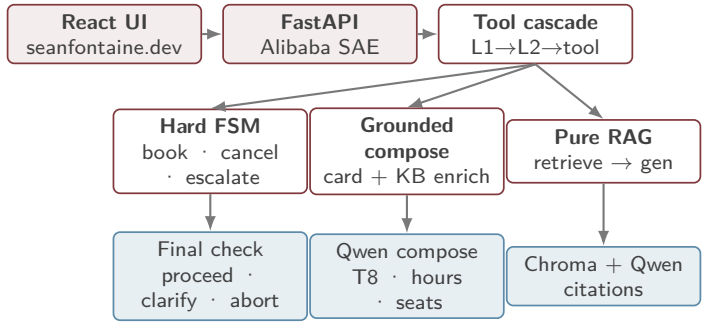
Tier	Examples	LLM / RAG
Hard FSM + final check	Book / cancel, escalate	Bypass · DST · cheap go-ahead
Live info + enrich	Typhoon+HKO, open now?, seats	Live doc + channel-pack enrich
Pure RAG	Fees, programmes, policies	Dense‖BM25 pack → generate

Detector awards (tools / live / RAG default) ≠ dense or BM25 chunk scores.

Final check returns `proceed` / `clarify` / `abort` (safety fail-opens).

Architecture

1



Inspector: match cue · score ledger · final check · grounding · RAG expand

What you will see live

2

Parent chat (WhatsApp-shaped)

- Messaging surface + probes
- Chat only — no booking form

System inspector

- Router decision + score ledger
- Final check · grounding · RAG expand
- Class register · session memory

Staff desk

- SheetPro follow-ups (gaps / assessment)
- MailBox urgent inbox
- Drafted hand-off emails

5-minute rule

Slides cue. We **operate** the system.

▶ **LIVE** Open the live demo fullscreen. Reset once.

Demo beats (cue card)

2

Beat	Show	Point at
1	Typhoon / open now?	Live info scores · Live→Pack enrich
2	Phonics fee	RAG default · Pack 3+2 · fee chunk
3	Seats → book Jason	DST · **Final check** · SheetPro
4	Scuba (unknown)	Honest gap · **SheetPro** row
5	Safety allergy call	Final check → Staff email → MailBox

▶ **LIVE** Operate the live system. Narrate the inspector — do not read this table.

Impact & why it works

3

Impact

- 24/7 triage without inventing policy
- Live tools for hours / typhoon / seats
- RAG for fees + programme narrative
- Humans only when stakes demand it

Creativity

- Channel pack + commit final check + DST
- Staff desk = Sheets + Gmail clones

Correctness cues

Score ledger + decision path
Citations with retrieval scores
Final check before irreversible writes
Urgent MailBox + drafted email

Evaluation

Live path correctness · visible RAG · latency feel.
No formal user study required.

Thank you

3 Impact

Grounded answers. Structured actions. Honest hand-offs.

RAG · tools · staff desk

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Q&A

Thank you for listening!
Your feedback will be highly appreciated!